Jonas Dann

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Research Interests

Hardware specialization and heterogeneous computer systems for large-scale data processing in the cloud. Current focus on SmartNICs for data lakes.

Education

Heidelberg University, *Computing Systems Group* 2019 – 2024
Ph.D., Computer Science (Graduation with highest honors: **Summa cum laude**)
Thesis: *FPGA-based Query Acceleration for Non-relational Databases*
Supervisor: Prof. Holger Fröning

Karlsruhe Institute of Technology 2016 – 2019
M.Sc., Computer Science
Thesis: *Improving Distributed External Sorting for Big Data in Thrill*
Supervisor: Prof. Peter Sanders

DHBW Mannheim 2013 – 2016
B.Sc., Applied Computer Science
Thesis: *Basic Components of Integration Systems on FPGAs*

Professional Experience

TU Darmstadt, Systems Group 2027
Requested as interim professor (Vertretungsprofessor) for the summer term 2027, invited by Prof. Carsten Binnig.

ETH Zürich, Systems Group 2024 – present
Postdoctoral Researcher
Research on hardware specialization and heterogeneous computing systems for large-scale data processing in the cloud.

SAP SE, HANA Database Campus 2019 – 2023
Industrial Ph.D. Candidate
Research on FPGA-accelerated query processing for graph and document stores.

SVA GmbH, Big Data & Analytics 2017 – 2019
Part-time Intern
Developed an Apache Spark-based batch analytics pipeline for a consulting client.

SAP SE, Sports Sponsoring – Sailing 2016 – 2017
Part-time Intern
Organized sailing events and developed analytics software for sailing regattas.

SAP SE 2013 – 2016
Cooperative Education Student
Dual study program with rotations across various departments.

Publications

[1] Jonas Dann, Luca Tagliavini, Gustavo Alonso. Oasis: Hiding the Cost of Querying Parquet Files in the Datapath. In *VLDB*, 2027.

[2] Jonas Dann, Gustavo Alonso, Yannis Chronis, Maximilian Heer, Sven Hepkema, Ana Klimovic, Vasilis Mageirakos, Benjamin Ramhorst, Silvia R. Alcaraz, Bowen Wu, Yu Zhu. CODAs: Computational Data Paths for Lakehouses. In *CIDR*, 2027.

[3] Silvia R. Alcaraz, Sven Hepkema, Vasilis Mageirakos, Jonas Dann, Gustavo Alonso. Analysis of Shared Memory Between CPUs and GPUs. In *DIMES @ SOSP*, 2026.

[4] Benjamin Ramhorst, Hongshi Tan, Luhao Liu, Silvia R. Alcaraz, Maximilian Heer, Geert Roks, Jonas Dann, Bingsheng He, Gustavo Alonso. Reinventing the Data Center FPGA. Under submission, 2026.

[5] Benjamin Ramhorst, Maximilian Heer, Luhao Liu, Heejae Kim, Jonas Dann, Jin-Soo Kim, Gustavo Alonso. SCENIC: Stream Computation-Enhanced SmartNIC. Under submission, 2026.

[6] Jonas Dann, Gustavo Alonso. Should I Hide My Duck in the Lake?. In *DaMoN @ SIGMOD*, 2026.

[7] Maximilian Heer, Benjamin Ramhorst, Yu Zhu, Luhao Liu, Zhiyi Hu, Jonas Dann, Gustavo Alonso. RoCE BALBOA: Service-enhanced Data Center RDMA for SmartNICs. In *OSDI*, 2026.

[8] Jiayong Li, Jonas Dann, Zhenhao He, Gustavo Alonso, Sai Rahul Chalamalasetti, Dejan Milojicic, Lance Evans, Alex Veprinsky, Runbin Shi. StreamDedup: Distributed In-line Deduplication for Disaggregated Storage. *TRETS, Volume 19, Issue 2*, 2026.

[9] Jonas Dann, Tobias Götz, Daniel Ritter, Jana Giceva, Holger Fröning, Gustavo Alonso. GraphMatch: Subgraph Query Processing on Steroids. In *SIGMOD*, 2026.

[10] Benjamin Ramhorst, Dario Korolija, Maximilian Heer, Jonas Dann, Luhao Liu, Gustavo Alonso. Coyote v2: Raising the Level of Abstraction for Data Center FPGAs. In *SOSP*, 2025.

[11] Marko Kabic, Bowen Wu, Jonas Dann, Gustavo Alonso. Powerful GPUs or Fast Interconnects: Analyzing Relational Workloads on Modern GPUs. In *VLDB*, 2025.

[12] Jonas Dann, Daniel Ritter, Holger Fröning. GraphScale: Scalable Processing on FPGAs for HBM and Large Graphs. *TRETS, Volume 17, Issue 2*, 2024.

[13] Jonas Dann, Daniel Ritter, Holger Fröning. Non-relational Databases on FPGAs: Survey, Design Decisions, Challenges. *CSUR, Volume 55, Issue 11*, 2023.

[14] Jonas Dann, Royden Wagner, Daniel Ritter, Christian Färber, Holger Fröning. PipeJSON: Parsing JSON at Line Speed on FPGAs. In *DaMoN @ SIGMOD*, 2022.

[15] Jonas Dann, Daniel Ritter, Holger Fröning. GraphScale: Scalable Bandwidth-efficient Graph Processing on FPGAs. In *FPL*, 2022.

[16] Jonas Dann, Daniel Ritter, Holger Fröning. Demystifying Memory Access Patterns of FPGA-based Graph Processing Accelerators. In *GRADES-NDA @ SIGMOD*, 2021.

[17] Jonas Dann, Daniel Ritter, Holger Fröning. Exploring Memory Access Patterns for Graph Processing Accelerators. In *BTW*, 2021.

[18] Daniel Ritter, Jonas Dann, Norman May, Stefanie Rinderle-Ma. Hardware Accelerated Application Integration Processing: Industry Paper. In *DEBS*, 2017.

Open Source Activity

Oasis, https://github.com/celeris-labs/oasis
SmartNIC for Parquet decoding and operator offloading in data lakes.

Coyote, https://github.com/fpgasystems/Coyote
FPGA shell providing operating system abstractions & FPGA virtualization.

libstf, https://github.com/fpgasystems/libstf
Standard library for data processing in hardware.

Patents

US Patent 18,053,505 2023
Parsing JSON on Field-programmable Gate Arrays
Jonas Dann, Daniel Ritter

US Patent 17,747,922 2023
Scalable Bandwidth-efficient Graph Processing on Field-programmable Gate Arrays
Jonas Dann, Daniel Ritter

US Patent 11,354,771 2022
Simulation Environment for Efficient Assessment of Memory-bound Platforms
Jonas Dann, Daniel Ritter

US Patent 10,176,146 2019
Integration Pattern Implementations Using Reconfigurable Logic Devices
Daniel Ritter, Jonas Dann

Teaching Experience

Teaching Assistant & Guest Lecturer Data Managememt Systems	2026
Head Teaching Assistant & Guest Lecturer Data Modeling and Databases	2026
Head Teaching Assistant & Guest Lecturer Data Management Systems	2025
Lecturer Computing Platforms Seminar	2025
Teaching Assistant & Guest Lecturer Data Modeling and Databases	2025
Teaching Assistant & Guest Lecturer Data Management Systems	2024
Guest Lecturer Data Modeling and Databases	2024

Mentoring

Research Assistants

Aaron Schüttpelz	2026

Master's Students

Luca Tagliavini, Loris Keist, Nicola Lohr, Andrei Girjoaba, Lukasz Wala, Matteo Fornaini, Stoyan Malinin, Vatsal Dixit, Anna Pecheanu	2026
Philipp Engljähringer, Rasmus Lüscher, Sven Weber, Sven Zanetti, Cedric Caspar, Filippo Selvatici	2025
Linus Vogel, Paolo Rondot, Shiduo Xin, Philipp Hardegger	2024
Felix Göken (Heidelberg University), Tobias Götz (TU Munich)	2022
Jan Ahlbrecht (TU Munich)	2021
Royden Wagner (Heidelberg University)	2020

Bachelor's Students

Bruce Röttgers, Janos Kreissl, Vincent Franz, Mehmet Emin Yaksi (Bilkent University), Yunus Emre Kök (Bilkent University), Simon Wieland, Simon Thalmann	2026
Sebastian Gavrilas, Jakob Klemm, Manuel Sandmeier	2025
Michael Egloff, Severin Obrist	2024

Community Service

Program Committee RAISE-Arch @ MICRO	2026
Reviewer VLDB Journal	2026
Reviewer VLDB	2026
Program Committee SoCC	2026
Program Committee HCDS @ ASPLOS	2026
Reviewer ICPP	2023
Reviewer Internationale Tagung Wirtschaftsinformatik (WI)	2023
Reviewer FPL	2021
Reviewer ICPP	2021

Talks

Oasis: Hiding the Cost of Querying Files in the Network Datapath

Invited Talk, Microsoft Gray Systems Lab, virtual	2026
Invited Talk, EPFL, Lausanne, Switzerland	2026
Invited Talk, CHORYS Workshop @ TU Dresden, Germany	2026
Invited Talk, University Heidelberg, Germany	2026
Invited Talk, TU Darmstadt, Germany	2026
Guest Lecture, TU Munich, Germany	2026

Building Stream Computation-Enhanced SmartNICs

Invited Talk, Stream2HPC @ ISC, Hamburg, Germany	2026

GraphMatch: Subgraph Query Processing on Steroids

Conference Talk & poster, SIGMOD, Bangalore, India	2026
Invited Talk, European Inno. Stars Workshop, Amsterdam, Netherlands	2024

Should I Hide My Duck in the Lake?

Workshop Talk & Poster, DaMoN @ SIGMOD, Bangalore, India	2026

Hardware Specialization for Data Analytics

Invited Talk, Oracle, US	2026
Invited Talk, AMD-Xilinx, US	2026
Invited Talk, Dell EMC, Germany	2025

Accelerating Modern Databases with FPGAs in the Datacenter

Invited Talk, European Innovation Stars Workshop, Brussels, Belgium	2023
Invited Talk, F4HD @ HiPEAC, Toulouse, France	2023

GraphScale: Scalable Bandwidth-Efficient Graph Processing on FPGAs

Conference Talk, FPL, Belfast, UK	2022

PipeJSON: Parsing JSON at Line Speed on FPGAs

Workshop Talk & Poster, DaMoN @ SIGMOD, Philadelphia, USA	2022

Demystifying Memory Access Patterns of FPGA-based Graph Processing Accelerators

Conference Talk, GRADES-NDA @ SIGMOD, virtual	2021

Exploring Memory Access Patterns for Graph Processing Accelerators

Conference Talk, BTW, virtual	2021